

MANIRATNAM MANDAL

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EDUCATION

Ph.D. in Electrical and Computer Engineering <i>The University of Texas at Austin</i>	2020 – 2024 GPA: 4.0/4.0
M.Tech. in Electrical Engineering <i>Indian Institute of Technology Kanpur, India</i>	2018 – 2019 GPA: 10.0/10.0
B.Tech. in Electrical Engineering <i>Indian Institute of Technology Kanpur, India</i>	2014 – 2018 GPA: 9.1/10.0
Foundation Masters – Signal, Control, and Robotics <i>Semester Exchange at École Centrale de Nantes, France</i>	2018 GPA: 10.0/10.0

AREAS OF INTEREST

- Image and Video Processing
- Computer Vision (CV)
- Computational Photography
- Vision Models
- Video Engineering
- Image & Video Quality Assessment
- Accessible Media Technology
- Data Science

RESEARCH AND WORK EXPERIENCE

GRADUATE RESEARCH ASSISTANT <i>Laboratory of Image and Video Engineering (LIVE), UT Austin</i> Supervisor: <i>Dr Alan C Bovik</i>	Jan'20 – Dec'24 Austin, Texas
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- **Text Legibility and Quality Assessment for UGC Media** 📺 (Sponsor: YouTube)
 - Created and analyzed **unique** subjective **datasets** for assessing text legibility and quality in visual media.
 - Developed **SOTA** deep learning **models** for text-in-image legibility and text-in-video quality prediction.
- **Subjective Portrait Region Cropping for UGC Video** 📺 (Sponsor: YouTube)
 - Developed the **largest** subjective **video cropping database**, advancing aspect ratio transformation research.
 - Conducted in-lab human study to analyze subjective video cropping techniques for portrait content creation.
- **Image Quality Assessment for Visually Impaired UGC** 📺 (Sponsor: Meta AI)
 - Created the **largest dataset** for image quality and distortion analysis for visually impaired UGC.
 - Developed a **multi-task scalable model** and real-time applications to provide quality and distortion feedback.
- **Video Quality Assessment for UGC Media** 📺 (Sponsor: Meta AI)
 - Created the **largest UGC** video quality database and conducted the largest subjective VQA study to date.
 - Developed a **SOTA blind video quality predictor** capable of generating spatial and temporal quality maps.

SUMMER RESEARCH INTERN <i>SMI Lab, Samsung Research America</i>	Jun'22 – Aug'22 Plano, Texas
• Designed an in-lab Video Quality of Experience Study involving real-time ratings for streaming applications. • Analyzed video quality metrics for assessing spatio-temporal anomalies generated during data transmission. • Assisted in modeling QoE anomalies prediction from Network Layer statistics.	

UNDERGRADUATE RESEARCHER <i>Computer Vision Laboratory, IIT Kanpur</i> Supervisor: <i>Dr K S Venkatesh</i>	Jul'18 – Jun'19 Kanpur, India
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- Proposed a novel method of **quasi-orthographic surface imaging** and analyzed it on generated topographies.
- Designed algorithms to compute imaging surfaces, derive bounds on imaging distance, and optimize capture points.
- Developed and compared sequential and batch filling algorithms for efficient surface reconstruction.

RESEARCH INTERN <i>Checko, IIT Kanpur</i>	Jul'18 - Sep'18 Kanpur, India
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- Developed 3D tag **counterfeit detection** pipeline based contrast gradients, reflection patterns, and print features.
- Implemented advanced **encryption** and **watermarking** techniques across Frequency, DCT, and Wavelet domains.

RELEVANT PUBLICATIONS

CONFERENCES

- M. Mandal, N. Birkbeck, B. Adsumilli, and A. C. Bovik, “*LegiT: Text Legibility for User-generated Media*,” IEEE International Conference on Image Processing (**IEEE ICIP**), **2024 (Oral Presentation)**. 
- C. Lee, M. Mandal, N. Birkbeck, Y. Wang, B. Adsumilli, and A. C. Bovik, “*Subjective Portrait Region Cropping on Landscape Video Study*,” IEEE International Conference on Image Processing (**IEEE ICIP**), **2024**. 
- Z. Ying, M. Mandal, D. Ghadiyaram, A. C. Bovik, “*Patch-VQ: ‘Patching up’ the Video Quality Problem*,” IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), **2021 (Oral Presentation)**. 

JOURNALS

- M. Mandal, N. Birkbeck, B. Adsumilli, and A. C. Bovik, “*Quality Prediction of Embedded and Overlaid Text in User-Generated Visual Content*,” IEEE Transactions on Image Processing (**IEEE TIP**), **2024** (under review).
- M. Mandal, D. Ghadiyaram, D. Gurari, and A. C. Bovik, “*Helping Visually Impaired People Take Better Quality Pictures*,” IEEE Transactions on Image Processing (**IEEE TIP**), vol. 32, pp. 3873–3884, **2023**. 

THESES

- Maniratnam Mandal, “*No-reference Image and Video Quality Assessment for User-generated Media*,” **Ph.D. Dissertation**, (The University of Texas at Austin, 2024). 
- Maniratnam Mandal, “*Optimum Methods for Quasi-Orthographic Surface Imaging*,” **M.Tech. Thesis**, (Indian Institute of Technology Kanpur, 2019). 

TEACHING EXPERIENCE

TEACHING ASSISTANT

2018 – 2019

Department of Electrical Engineering, IIT Kanpur

- Worked as a TA for courses – **Introduction to Electronics**, **Image Processing**, and **Solid State Devices**.
- Responsible for assistance in course tutorials, guidance in course projects, and grading assignments and exams.

VOLUNTEER MEMBER

2018 – 2020

English Proficiency Program, IIT Kanpur

- Management of the courses – **English Proficiency and Scholarly Communication** and **Practical English: Learning and Teaching** (online).
- Responsible for creating video lectures, and crafting and grading of assignments.

AWARDS

- Awarded **Outstanding TA Award** by Dept. of Electrical Engineering, IIT Kanpur, **2019**.
- Awarded **Academic Excellence Award** by IIT Kanpur for **consecutive academic years 2014 – 18**.
- Recipient of **Erasmus+ EU scholarship** for semester exchange in France, **2018**.
- Awarded **Kishor Vaigyanik Protsahan Yojana (KVPY)** scholarship by DST, Govt. of India, **2013**.

SELECTED COURSEWORK

ENGINEERING	Digital Video; Image Processing; Advanced Computer Vision; Vision Systems; Computer Programming and Data Analysis; Data Mining; Embedded and Cyber-physical Systems; Statistical Machine Learning; Digital Signal Processing; Speech Signal Processing
MATHEMATICS	Probability and Statistics; Linear Algebra; Complex Analysis and Differential Equations; Convex Optimization; Statistical Signal Processing
CERTIFICATIONS	Machine Learning  (Stanford Online); Introduction to Neural Networks and PyTorch  (IBM); Deep Learning Specialization  (Deeplearning.ai); TensorFlow Developer Specialization  (Deeplearning.ai); Generative Adversarial Networks (GANs) Specialization  (Deeplearning.ai); Introduction to Image Generation  (Google)

TECHNICAL SKILLS

PROGRAMMING	Python, MATLAB, Javascript, HTML, CSS, Git, Bash, L ^A T _E X
TOOLS	TensorFlow, PyTorch, Scikit-Learn, Scikit-image, Scikit-video, OpenCV, Darts

ADDITIONAL PROJECTS

KAGGLE COMPETITIONS

- ❑ **Store Sales – Time Series Forecasting**  **2024**
 - Ranked in **top ten** out of 762 teams with a Root Mean Squared Logarithmic **Error of 0.378**.
 - Built and optimized **ensemble models** using **XGBoost**, **LightGBM**, and **TiDE**, leveraging advanced time series techniques, including past covariates, future covariates, and lag features, to improve forecasting accuracy.
- ❑ **Intruder Detection through Webpage Session Tracking**  **2023**
 - Boosted **F1 score** to **0.91** in web-user identification by addressing imbalance with downsampling and weighting.
 - Ranked in **top fifty** out of 5580 teams, achieving **ROC AUC of 0.969**, using **Light AutoML** to train LightGBM, LR, and XGBoost ensembles.
- ❑ **Predicting Domestic Flight Delays**  **2022**
 - Ranked **first** on the leaderboard, achieving **ROC AUC of 0.959**, using a tuned **CatBoost** model.
 - Leveraged Random Forest to generate new features for **binary classification**, improving AUC in a highly imbalanced dataset by utilizing class weights and implementing a custom focal loss function.

COMPUTER VISION

- ❑ **Combining Compression Techniques for Computer Vision**  **2021**
 - Compared compression efficacy of **quantization**, **pruning**, and **knowledge distillation** for smaller networks.
 - Analyzed the combination of different techniques when applied partially and sequentially to deep networks.
- ❑ **Improving Defensive Distillation using Teacher Assistant**  **2021**
 - Evaluated **distilled models** for different distillation temperatures in terms of accuracy, sensitivity, and robustness.
 - Demonstrated that multi-step distillation improves **robustness against adversarial attacks** in most cases.
- ❑ **Foreground Detection and Background Separation in Videos**  **2020**
 - Explored algorithms based on **PCA**, **GMM**, and **foreground motion estimation** for detecting moving subjects.
 - Developed algorithms based on motion estimation for removing moving objects to extract the static background.
- ❑ **Deep-fake Image and Video Detection Techniques**  **2020**
 - Surveyed different categories of **facial manipulation** in videos and images, and databases available for research.
 - Analyzed both classical and learning-based popular Deep-fake detection techniques.

GENERAL

- ❑ **COVID-19: Impacts and Insights**  **2020**
 - Modeled death and case projections in target demographics using **time-series analysis** and **LSTMs**.
 - Applied ML techniques to analyze and predict the impact of the pandemic on mental health and well-being.
- ❑ **Comparison of HRTF Pre-processing Techniques**  **2019**
 - Investigated the **perceptual effect** of Head Related Transfer Function (HRTF) pre-processing techniques.
 - Demonstrated using **energy analysis** that lower order SHT coefficients can be used for faster reconstruction.
- ❑ **Robust Optimization in Logistics**  **2019**
 - Studied the **retailer-supplier flexible commitment (RSFC)** problem to manage the supply chain logistics.
 - Implemented and analyzed three algorithms for optimizing the parameters based on uncertainty in demand.
- ❑ **Online Reconstruction from Big Data via Compressive Censoring**  **2018**
 - Studied **Sparsity-aware Censored Maximum Likelihood Estimator (SC-MLE)**, and tested the performance and convergence of the proposed optimization algorithm on sparse online data for real-time processing.

POSITIONS OF RESPONSIBILITY

- **INTERNSHIP AND COMPANY COORDINATOR (2015-16):** Managed college placement procedures and involved in creating preparation guides as a member of the **Student Placement Office, IIT Kanpur**.
- **STUDENT GUIDE AND ACADEMIC MENTOR (2015-16):** Member of the **Counselling Service, IIT Kanpur**, responsible for the orientation, and academic and personal guidance of the undergraduate freshmen batch.

REFERENCES

Up to three references available on request